

Addressing the Technology Learning Divide Using Co-Learning with Familiarization Method

Anik Saha

Design Inclusion and Access Lab,
Department of Electrical and
Computer Engineering, North South
University
Dhaka, Bangladesh
aniksaha@northsouth.edu

Naimur Rahman

Design Inclusion and Access Lab,
Department of Electrical and
Computer Engineering, North South
University
Dhaka, Bangladesh
naimur.rahman2@northsouth.edu

Nova Ahmed

Design Inclusion and Access Lab,
Department of Electrical and
Computer Engineering, North South
University
Dhaka, Bangladesh
nova.ahmed@northsouth.edu

ABSTRACT

Empowerment of low-literate women and marginalized communities through learning technology is emphasized by researchers. Co-learning techniques using familiarization methods can effectively help low-tech literate people learn technology, which is underexplored. This pilot study was conducted in two phases. Phase-I explored technology's learning and adaptation strategies with a qualitative study, where the researcher considered N=36 female participants in urban and rural regions of 3 major divisions of Bangladesh. Phase-II was conducted N=16 (Parent=8, Children=8) participants in Dhaka division urban and rural settings with a series of workshops to understand the impact, opportunities, and challenges of co-learning and familiarization methods in terms of learning technology. Findings show that technology co-learning with familiarization methods greatly aids low-literate women in adopting and pursuing technology rapidly. The findings are essential to provide insights to the CHI, and CSCW community regarding opinions on technology learning and adaptation scenarios of low-tech literate women in developing countries perspectives.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Familiarization, Co-learning, Technology Learning, Tech Adoption, Marginal Community

ACM Reference Format:

Anik Saha, Naimur Rahman, and Nova Ahmed. 2024. Addressing the Technology Learning Divide Using Co-Learning with Familiarization Method. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3650969>

1 INTRODUCTION

Recent studies have shown that women and marginalized communities in Bangladesh are far behind in technology usage and

learning [29–31]. Many barriers women face to explore technology are social challenges such as patriarchal practices [33]; while studies show enthusiasm and empowerment through technology learning [26]. Co-learning techniques, where a varied group of participants learns together, show opportunities for learning and knowledge exchange [15], and involve continuous sharing among the learners [17]; particularly learning about technology [28]. Previous studies explored that co-learning enhances the learning process with better and quicker output and also integrates peers in learning, helping users familiarize themselves with potential scenarios [32]. Close acquaintances help to acquire much information too[9]. The learning can be strengthened further using the familiarization method that helps participants to introduce new elements and objects [4]. The familiarization method is an approach of making familiar research elements with research objects. Familiarization helps to build researcher-participant rapport. According to Barley and Bath, familiarization is very effective for special groups that are very hard to reach (e.g., low-literate, people with disabilities, people with autism, etc.) [4]. Our research explores the concept of opening up co-learning opportunities through familiarization techniques for low-tech literate and low-literate women for technology exploration and learning through a small-scale pilot study.

Low-literate users mostly struggle with access, usage, and learning of technology interfaces [20], and recent research has explored numerous factors that work as hindrances to the learning and adoption of smartphones among low-literate users in developing countries, particularly in South Asia. Complex user interfaces containing a lot of text, difficult-to-access hierarchical menus, and lower availability of touchscreen devices pose challenges for low-literate users in learning new technology [11, 18]. Moreover, because of less capacity of bearing and working memory, older adults can bear less information in their brain and they have less confidence in using technology [11, 18]. However, prominent researchers have explored the concept of introducing technology learning opportunities. For instance, Medhi et al. highlighted that low-literate and novice technology users from developing countries adopt technology quickly when they can relate the design with familiar interfaces [20]. Similarly, low-literate taxi drivers from India could quickly adopt the researchers' design as they were familiar with the meaning of red or green coding, which was common to other paper resources they had used previously [35]. These research works focused on a specific group of people where there is a research gap regarding technology learning scenarios, opportunities, and challenges for participants with low-tech literacy in a developing country. Our work aligns

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650969>

with the existing body of literature exploring learning barriers and opportunities for low-literate and low-tech literate women.

Recent research has shown that technology learning and adoption create opportunities (e.g., developing and learning skills using technology, continuation of business through online, emergency support through online social communication) for the women of marginalized communities in Bangladesh [26]. In the context of Bangladesh, financial technology (fintech) adoption helps low-income communities to improve their financial condition [14]. As technology adoption and learning, positively impact regular life, there are some challenges to adopting and learning technology for low-literate and low-income people, particularly women. Low-literate participants often do not understand the application interfaces [20], so the familiarization phase may alleviate the above issues [4]. According to a recent study in Bangladesh, low-income communities look for familiar technology designs to learn and adopt [35]. However, technology adoption and knowledge help low-literate women to increase their confidence to participate in family decisions [5] and their interest in learning technology has a positive impact on their regular life [25]. While the existing research works discussed technology adoption barriers and opportunities from the perspective of developing countries, technology learning and adoption strategies of low-literate women are still unexplored. This research gap motivates the researchers to use familiarization methods along with co-learning workshops to understand technology learning and adoption strategies of low-literate women in developing countries.

The aim of this research includes the understanding of the learning challenges and opportunities associated with the aforementioned methods. This research is also important to discuss how technology adoption can be accessible for low-literate women in developing countries. Moreover, the following research question is explored in the context of developing countries:

- RQ: How co-learning with familiarization method facilitated low-tech literate women in technology learning?

This research has been conducted in 2 phases, in phase I, we conducted a qualitative study on N=36 in rural and urban settings of 3 major divisions of Bangladesh, and in Phase II we conducted a series of workshops on N=16 (parent=8, children=8) in the rural and urban areas of Dhaka division. Phase I was closely studied to understand participants' technology learning and adoption strategies, and learning challenges from the participant's point of view, and this research explored that co-learning works very well for this community. Phase II was conducted to understand the impact of co-learning and familiarization in terms of technology learning. The research revealed that co-learning and familiarization work as a catalyst for learning technology in the context of developing country perspectives. The study findings motivated us to diversify the CHI, and CSCW community's understanding by discussing how technology learning occurs and can be enhanced with familiarization and co-learning workshops for low-literate communities in developing countries.

2 RESEARCH METHODS

The study took place through two phases; in Phase I, qualitative research took place to understand the barriers of low-literate and

illiterate participants in accessing and learning technology, and in Phase II the co-learning and the familiarization method were explored with a series of workshops. Phase I of the study was conducted between December 2020 and May 2022 when the COVID-19 pandemic was present, and Phase II of the study was ongoing from March 2023. In phase I the researcher interviewed N=36 female participants from urban and rural areas of Dhaka, Rajshahi, and Sylhet divisions, while in phase II N=16 (Parent = 8, children = 8) participants joined in a series of workshops from urban and rural settings of Dhaka division. All discussions were in Bengali which is the native language of the participants and researchers.

2.1 Participant Recruitment

In phase I, the participants were approached through purposive and convenient sampling, assuring that each focus group included people from homogenous groups (e.g., similar professions, and socioeconomic backgrounds who had common experiences). Participant demographics of phase I are shown in Table 1 where participants' income classifications are based on Pew Research [24]. Later in phase II, the participants were recruited through a purposive sampling method for workshops. During participant recruitment, the researchers contacted a locally trusted key informant (KI) to find homogenous groups for each interview. Before the focus group interviews and workshops, the researchers gathered information about participants' demographics, living areas, and local norms through discussions with the KI. The KI had also explicitly stated the research purpose to the participants before the interviews and workshops. In Phase I, we denoted participants with P and in Phase II we denoted parent participants with P and children participants with C.

2.2 Procedure of Interviews and Workshops

Phase I: In total 7 Focus group interviews with N=36 (All female) participants took place in-person in both urban and rural areas of 3 divisions of Bangladesh. The KIs confirmed the researcher's interview mode, time, and places based on the participants' preferences, which the researchers followed. For the focus group discussion, the group size consisted of 4 to 5 participants as preferred by Kruger [16]. Researchers described the topic to participants and asked about their choices, worries, or reservations regarding their technology learning process and barriers. The researchers followed a semi-structured questionnaire for this phase. Each interview took nearly 60-150 minutes. The moderator used the introduction session as an opportunity to be familiar with the participants. One researcher who had experience handling challenges like dominance in a discussion moderated the interview. The moderator started the discussion initially and the remaining researchers took notes and helped in asking follow-up questions. The researcher provided face masks and hand sanitizer during the in-person interviews to maintain the World Health Organization (WHO) protocols for safety precautions.

Phase II: A total number of 8 workshops (co-learning along with familiarization method) were conducted with N=16 (Parent=8, Children=8) participants in this phase. Two groups of participants, each group contained 4 parent participants and 4 children participants from both urban and rural areas of the Dhaka division. Each group

Table 1: Study group demographic of Phase I.

Group Name (Gender, Number)	Region (Division)	Education Level	Income Level	Use Phone Type
Ready-made Garments Worker (Female, 5)	Dhaka (Urban)	No Education (5)	Low Income(5)	Feature Phone (5)
Tant Weaver (Female, 5)	Dhaka (Rural)	No Education (5)	Low Income (5)	Feature Phone (3) Smart Phone (1) No Phone (1)
Potters (Female, 5)	Dhaka (Rural)	No Education (5)	Low Income (5)	Feature Phone (5)
Silk Mill Worker (Female, 5)	Rajshahi (Urban)	No Education (5)	Low Income (5)	Feature Phone (5)
Paper Packet Makers (Female, 5)	Rajshahi (Urban)	No Education (4) SSC (1)	Low Income (5)	Feature Phone (3) No Phone (2)
Rice Mill Worker (Female, 5)	Rajshahi (Rural)	No Education (5)	Poor (5)	Feature Phone (3) No Phone (2)
Tea Garden Workers (Female, 6)	Sylhet (Rural)	No Education (5) SSC (1)	Poor (6)	Feature Phone (3) No Phone (3)

participated in 4 consecutive workshops that were held once a week. The length of each workshop was nearly 90-150 minutes. In the first workshop of each group, the familiarization method was implemented to make participants familiar with research elements and build relationships between researchers and participants. The researcher used the familiarization method to establish their positionality and familiarize the participants with smartphone interfaces and some features they identified to learn (e.g., Make Call, Facebook, Messenger, YouTube, etc.). In the familiarization phase, the researcher began by introducing himself as a researcher who lacks knowledge about low-literate and low-tech literate women's technology adaptation and wants to help them learn technology uses that they wanted to learn. The researchers presented a session about how smartphones differ from feature phones and how to operate them. Later they asked each individual what kind of application they wanted to learn and took note of their phone-using skills. The researchers used to live in the same village during the workshops so that they can make the relationship easy and trustworthy which helped to get quality data and information. Additionally, the researchers could know about their lifestyle and technology usage behavior in the process. The technology co-learning workshops were implemented in the rest of the 3 workshops. Researchers identified some common applications that the participants wished to learn and presented how every feature of the applications works and how to operate them. Finally, the researcher showed every application's features, steps, and skills for operating them several times until it became easy for them. A short focus group discussion session that lasted for 20-30 minutes about the learning progress and challenges during the week before each workshop started from the second co-learning workshop. In the Technology co-learning workshops, the researcher connected the child with their mother as a technology teaching mentor as children have better knowledge and can aid their parents if they face any difficulties while learning

as shown in Figure 1. Children mostly acted as helping hand to their mothers on how to operate smartphones and their desired applications (e.g., Facebook, YouTube, Video Call) in their free time.

2.3 Qualitative Content Analysis

After the first interview, the informal analysis process started and it continued till the last interview. Audio recordings of interviews were transcribed, anonymized, and translated to English. The researchers imported the transcriptions into Atlas.ti Cloud [2] and started qualitative coding. During the pilot study, the open coding phase started [8], researchers read the transcripts line by line several times. After every interview, researchers sat together and discussed the interview findings and created summary notes. In total, researchers identified 37 codes in the open coding phase. We contrasted and combined these initial codes during the axial coding stage to identify the high-level themes. The researchers created a schematic diagram of codes during the study and continued refining it throughout the analysis process using Miro [21] to build relationships among the axial codes. In any disagreement, they discussed it among themselves and reached a consensus.

2.4 Research Ethics and Incentives

The Institutional Review Board authorized this study. The researchers outlined the research aims, principles, discussion techniques, guidelines, and permissions of recording the discussion, as well as answering participants' questions before the discussions, where participants might quit the conversation at any moment without ceding their incentives. All notes, information, recordings, and write-ups are safely kept in a protected drive to which only the researchers have access. All the participants got BDT 1000 (1 USD=106.06 BDT) as incentives.

2.5 Positionality

All of the researchers were born and brought up in Bangladesh. The research was conducted by mixed-gender researchers from different socioeconomic backgrounds and religions who have several years of qualitative research experience in Bangladesh on technology. The researchers' backgrounds and positions are disclosed based on Ruth Franekber's results, which state that it is essential for readers to understand the perspectives utilized [10].

3 PHASE I: EXPLORING BARRIERS AND OPPORTUNITIES IN TECHNOLOGY LEARNING

Several participants from Phase I of the study shared about basic technology operations such as making phone calls, saving contact numbers, sending and receiving messages - particularly, messages related to aid and stipends during the pandemic period. A group of low-income participants such as the rice mill workers or tea garden workers shared their inability to maintain a mobile phone considering their low income level and responsibilities to run their families. A rice mill worker mentioned how mobile phones do not make sense when they do not have enough food to fill their stomach. Similarly, a tea garden worker considered mobile phones to increase their daily cost as a reason behind not owning a phone for herself. It was observed that the tea garden region has a poor telecommunications network and making a proper call requires leaving the garden area. The participants shared about their laborious days when there was very little time left to explore technology learning. Consideration of entertainment using technology appeared as a complex problem for many of the participants of consideration.

The advanced applications participants discussed included financial applications (e.g., mobile banking, mobile wallets), social media usage, and creating custom video playlists from YouTube or other similar platforms that require internet connectivity. The financial application category was an important one to ensure contactless financial transactions during the COVID-19 pandemic period. The participants shared their eagerness to learn about mobile wallets, particularly, the few that are widely used in Bangladesh. The participants working in the ready-made garment (RMG) sector shared about being paid through their mobile wallets and accessing the money through the agents. The participants shared that the majority of them were unable to comprehend the existing interfaces which acted as barriers to adopting mobile wallet accounts. A participant shared how she tried to learn the steps of sending money over and over but forgot the steps eventually. A few of them could only read simple sentences in their native language, therefore they faced problems in operating fintech applications. They used their phones mostly for making phone calls by recalling numbers and symbols. Participant P9, a weaver, shared her difficulties with understanding SMS. She sought her daughter's assistance to read messages and her nephew's help to withdraw money. The struggle is shared as follows:

"I use big phones. I do not know much about these things. My daughter understands all these things. She informs me if the money comes in. When money comes on my mobile, then I call my nephew, and

he take me to the agent shop." -P9, Weaver, Female, Rural Area, Tangail.

The challenges were beyond using the advanced applications that need multiple steps to complete, it was related to the lack of opportunities to learn the applications. An RMG worker shared how she was determined to learn to send money to her family all by herself but she would forget the steps in between. Members of the marginal communities struggle to access adequate learning opportunities.

However, a previous study in Bangladesh [1], explored that the launch male workers were able to learn and take full advantage of technology despite being low-literate as they had their peers and colleagues to learn from. It shows that opportunities to learn a technology can significantly vary depending on gender and exposure to different communities which makes co-learning for female low-literate and low-tech literate participants more essential.

4 PHASE II: EXPLORING CO-LEARNING AND FAMILIARIZATION WORKSHOP

In this section, we discussed the opportunities and advantages of co-learning and familiarization workshops. Table 2 illustrates the outcomes of the workshops.

4.1 Passion vs Limitation

In this study several participants shared that they want to learn smartphones but some limitations play a negative role in learning. Participants shared about complicated user interfaces, time limitations, having no mentor, lack of memorizing skills, age barriers, having helping hands for assistance, etc. as barriers in learning technology. The participants shared their eagerness to learn more being aware of existing limitations. According to P4, *"I tried a little bit to learn how to operate a big phone (smartphone), but it looks complicated"*. Another participant P3 shared that she did not get enough time to operate her smartphone as she had other work to do. According to the participants, having children or spouses who can operate smartphones for them enabled them to ignore technology learning as they could delegate them. P1 shared, *"I don't need to do a video call or YouTube browsing, I have my children and they do it for me"*. On the other hand complicated user interface and low memorizing skills hinder participants learning, as shared by P2 are given below:

"The interface and features seem complicated, and I cannot remember them in my mind for a long time." - P6, Housewife, Female, Urban Area, Dhaka.

Another participant, P1, shared the same sentiment, expressing that she can bear less information in her brain due to her age. Despite facing numerous limitations, they remain passionate about learning technology and being independent

4.2 Children as a Savior

In this research, we explored that co-learning works as a catalyst for parents with their children in learning technology. Children who are familiar with the technology can help their guardians achieve smooth and quick learning results as shared by the participants. One of the participants P1 mentioned that she has a lot of work to

Table 2: Outcomes of the Workshops

Participants, Gender, Region, Education	Skills Before the Workshop	Expectations of Learning from the Workshop	Skills After Completion of the Work- shop
P1, Female, Ru- ral, Graduate	Feature Phone: Make call, Receive Call, Save Number Smartphone: Nothing	Smartphone Operation, Facebook, Video Calls, YouTube	Smartphone: Call, Receive Call, YouTube, Facebook, Video Call
P2, Female, Ru- ral, Class 8	Feature Phone: Make call Receive Call. Smartphone: Nothing	Smartphone Operation, Facebook, Video Calls, YouTube	Smartphone: YouTube, Facebook, Video Call
P3, Female, Ru- ral, SSC	Feature Phone: Make call Receive Call. Smartphone: Nothing	Smartphone Operation, Facebook, Video Calls, YouTube	Smartphone: Call, Receive Call, YouTube, Facebook, Video Call
P4, Female, Ru- ral, HSC	Feature Phone: Make call Receive Call. Smartphone: Nothing	Smartphone Operation, Facebook, Video Calls, YouTube	Smartphone: Call, Receive Call, YouTube, Facebook, Video Call, Camera
C1, Male, Rural, Class 9	Feature Phone: Call, Receive Call, Save Number Smartphone: Call, Receive Call, YouTube, Video Call	Full functionality of Facebook, E- commerce, Online Payment	Online Payment, E-commerce shop- ping, Facebook functionality
C2, Male, Rural, Class 11	Feature Phone: Call, Receive Call, Save Number Smartphone: Call, Receive Call, YouTube, Video Call	Full functionality of Facebook, E- commerce, Online Payment	Online Payment, E-commerce shop- ping, Facebook functionality
C3, Female, Ru- ral, Class 8	Feature Phone: Call, Receive Call, Save Number Smartphone: Call, Receive Call, YouTube, Video Call	Full functionality of Facebook, E- commerce, Online Payment	Online Payment, E-commerce shop- ping, Facebook functionality
C4, Female, Ru- ral, Class 7	Feature Phone: Call, Receive Call, Save Number Smartphone: Call, Receive Call, YouTube, Video Call	Full functionality of Facebook, E- commerce, Online Payment	Online Payment, E-commerce shop- ping, Facebook functionality
P5, Female, Ur- ban, SSC	Feature Phone: Make call, Receive Call, Save Number Smartphone: Facebook Scrolling, YouTube scrolling	Smartphone Operation, Facebook, Video Calls, Google, MFS: bKash	Smartphone: Save Number, YouTube search by voice and words, Facebook, Video Call, Capture Photos, MFS: bKash
P6, Female, Ur- ban, Class 8	Feature Phone: Make call, Receive Call Smartphone: Only Scroll YouTube Videos	Smartphone Operation, Facebook, Video Calls, YouTube, bKash	Smartphone: Save Number, YouTube search by voice, Facebook, Video Call, Capture Photos, MFS: bKash
P7, Female, Ur- ban, Class 5	Feature Phone: Make call, Receive Call Smartphone: Nothing	Smartphone Operation, Facebook, Video Calls, YouTube, Others app like bKash	Smartphone: Save Number, YouTube search by voice, Facebook, Video Call, Capture Photos, MFS: bKash
P8, Female, Ur- ban, Class 7	Feature Phone: Call, Receive Call Smart- phone: Scroll Facebook	Smartphone Operation, Facebook, Video Calls, YouTube	Smartphone: Save Number, YouTube search by voice, Facebook, Video Call, Capture Photos, MFS: bKash
C5, Female, Ur- ban, Class 10	Feature Phone: Make call, Receive Call, Save Number, send Messages Smart- phone: Call, Receive Call, Scrolling Face- book and YouTube	Full functionality of Google, E- commerce, Online Payment	Full functionality of Online Payment, E-commerce shopping, Facebook func- tionality
C6, Female, Ur- ban, Class 11	Feature Phone: Make call, Receive Call, Save Number Smartphone: Call, Receive Call, YouTube.	Facebook, E-commerce, Online Pay- ment	Online Payment, E-commerce shop- ping, Facebook
C7, Male, Ur- ban, Class 8	Feature Phone: Make call, Receive Call, Save Number Smartphone: Call, Receive Call, Video Call.	Facebook, E-commerce, Online Pay- ment	Online Payment, E-commerce shop- ping, Facebook functionality, YouTube
C8, Male, Ur- ban, Class 11	Feature Phone: Make call, Receive Call, Save Number Smartphone: Call, Receive Call, YouTube, Video Call	Full functionality of Facebook, E- commerce, Online Payment	Online Payment, E-commerce shop- ping, Facebook functionality



Figure 1: Co-learning Activity: Children showing their parents based on parent's requests

do and can't get enough time to learn to operate smartphones by herself as it takes time to become familiar, but when her son started helping her it impacted her positively. As shared by participant P2, *"I got very little time to learn because of my pottery work, but it wouldn't be possible if the children didn't help me to learn"*. Along with it, they mentioned that children are enthusiastic as mentors who can go over a topic multiple times as required. As shared by participant P4, *"Children don't get bored while they teach us. They are very enthusiastic"*.

4.3 Breaking Technology Dependency

We found that most of the participants were dependent on their children or spouses while operating smartphones and browsing their desired content. Participants need to request or push their children or spouses while taking technological support from them. Now, after participating series of co-learning workshops with their children, they have become self-dependent as we have observed in our study. Participant P1 mentioned, *"Now I can watch whatever I want to watch on YouTube by myself. Before I needed to push others to run my desired videos, Now I can do it by myself"*. Participant P2 mentioned that she doesn't have any dependency after learning how to browse YouTube, now she can enjoy YouTube videos when she is having food and she really loves it. After learning to make video calls with help from their children now they can talk with their desired person via smartphones. Participant P3 shared that she wanted to talk with her brother who lives abroad by video call, but she needed to seek help before learning how to make video calls, which doesn't always work as her spouse and children have their work. Now she can make video calls by herself and talk with her siblings whenever she wants, she doesn't need to seek help from others. We also explored in this study that self-dependency opens a new horizon for the participants. They have started learning new things and skills with the help of YouTube videos. Now they can enjoy their desired videos in smartphones by themselves, as shared by the participant P4 are given below:

"Now I can watch my favorite song on YouTube, I also watch new recipes and learn some new items. When I didn't know how to run YouTube videos at that time, I needed to request someone to play videos for me and it took time, now I can do it by myself" -P8, Housewife, Female, Urban Area, Dhaka.

Co-learning workshops with the familiarization method helped the participants to become self-dependent on operating smartphones as we observed in our study.

5 DISCUSSION

Our study explored the importance of learning facilities for women in phase I of the study, which was validated in phase II through creating co-learning with familiarization techniques. Our study aligns with the existing body of research from the South Asian context. While other research focuses on various challenges women and marginal communities face in terms of technology access, learning, and adaptation; our work focuses on a possible solution approach to this problem.

Our study found a practice that creates technology learning facilities within their existing constraints. Many of the participants took advantage of their close-knit families and acquaintances for technology learning using co-learning and familiarization techniques. It shows the value of the collectivist social structure [12, 36], which can add value through collaborative learning facilities. Previous studies have shown that technology learning opportunities are constrained for women and marginalized communities in the South Asian Context for various social regulations [30, 33]. The problem is further complicated for rural women who often need to comply with patriarchal barriers such as obtaining permission for technology usage from their in-laws [33]. There are supportive male family members often who take care of the technology matters of the women in the households [27], which only increases the technology dependency for women, a problem that has been stated

as supportive masculinity [7]. The co-learning facility on the other hand, can offer technology independence for women and marginal community users.

The low-literate communities struggle with technology when the operations of a particular technology have multiple steps as in the case of fintech applications seen in our study. There are important works that addresses such problems through non-text and spoken-based application design [19], familiar design interfaces [20], simple technology design [6, 22, 34], contextualized application design [13, 23] and enhancement of technology literacy and technology education [14]. These problems require the attention of the application designers, authorities, and policymakers to ensure improved design opportunities, ensuring technology literacy etc. Our work complements these existing efforts and offers immediate remedies that the participants have used themselves to learn technology. A combination of enhanced application design, policy to ensure great technology exposure and co-learning facility can create great opportunities in the coming days.

The case of supporting to learn technology explored in this research differs from the ways discussed by Sambasivan et al. [30]. In this research, it is a collaborative learning process where Sambasivan et al. referred to technology dominance by men. Alternatively, we have identified that adopting technology is difficult for some participants due to a lack of understanding of the technology interface because of low literacy and confidence, lack of support in learning, language barrier, and lack of access to the phone. These findings fill the gaps of previous research that highlighted the limitation of learning technology for interface-related issues and education level only [25].

To learn technology quickly under any circumstance, especially contextual and user-centered technology, familiarization methods and co-learning with family members and close acquaintances are essential.

6 CONCLUSION AND WAY FORWARD

Women and marginalized communities in Bangladesh usually do not receive adequate technology learning opportunities as per our study. This research focuses on a pilot development of co-learning facilities with familiarization methods to help the participants to become self-sufficient in operating smartphones. It is proof of a concept pilot study suggested by the participants who initially shared their struggles with learning and using technology. A socially comfortable learning space to learn and practice smartphone usage opens up opportunities and self-reliance, and that is important for the empowerment of women in this region.

In this pilot study, the selected female participants were paired with their children who learned about smartphone usage as parent and children teams in a co-learning environment. The participants participated in a series of workshops that were important to make sure they were familiarized with the technology. The study allowed co-learning among parent-children pairs that were shared as an effective technique for technology learning from some of the participants. The co-learning technique was positively received by the participants, particularly, the female matured participants learning from their children. The removal of dependency on others was shared to be liberating by different female participants. It allowed

a comfortable learning space without any social prejudice where participants as parents were able to share their desire to learn about specific things (e.g., wanting to learn how to search for a favorite song on YouTube, being able to make a video call to a relative living abroad) and try them out on their own. The agency the female participants received was important as it showed a form of empowerment as seen in recent studies [26], being aware of the technology divide of women in this region [29–31]. A co-learning environment using the familiarization method shows great opportunity for the greater body of research in the field of feminist HCI that focuses on ensuring inclusive technology access, specifically, for women and marginalized communities [3, 4, 15, 17, 28].

This pilot study on technology learning facilities holds promise for future technology learners. The researchers hope to expand this work beyond the regional pilot study over a longer period of time to ensure a sustainable technology learning process specifically, for women and marginalized communities.

REFERENCES

- [1] Nova Ahmed, Anik Saha, Sabbir Ahmed, Ifthi Azad Abeer, Anik Sinha, Rahat Jahangir Rony, Syeda Shabnam Khan, Shajnush Amir, Tanvir Hasan, and Gazi Fakhrul Islam. 2022. Fintech as a Positive Enabler During COVID-19 in Bangladesh. *Journal of Bangladesh Studies* 24, 2 (2022), 16–24.
- [2] Atlas.ti. 2023. *Atlas.ti*. Atlas.ti. Retrieved February 3, 2023 from <http://web.atlasi.com>
- [3] Shaowen Bardzell and Jeffrey Bardzell. 2011. Towards a feminist HCI methodology: social science, feminism, and HCI. In *Proceedings of the SIGCHI conference on human factors in computing systems*. Association for Computing Machinery, New York, NY, United States, Vancouver BC Canada, 675–684.
- [4] Ruth Barley and Caroline Bath. 2014. The importance of familiarisation when doing research with young children. *Ethnography and Education* 9, 2 (2014), 182–195.
- [5] Monisha Biswas, Misita Anwar, Manika Saha, Nova Ahmed, Yolande Strengers, Larry Stillman, and Gillian Oliver. 2022. The World is in My Hand Now: Smartphones for Empowering Rural Women in Developing Countries: Smartphones for Empowering Rural Women in Developing Countries. In *Proceedings of the 2022 International Conference on Information and Communication Technologies and Development*. Association for Computing Machinery, New York, NY, United States, Seattle, USA, 1–11.
- [6] Gregory Chen, Alexandra Fiorillo, and Michel Hanouch. 2016. *Smartphones and Mobile Money: Principles for UI/UX Design (1.0)*. CGAP. Retrieved January 5, 2023 from <http://www.cgap.org/research/slide-deck/smartphones-mobile-money-principles-uiux-design-10>
- [7] Radhika Chopra. 2003. Rethinking pro-feminism: men, work and family in India. In *Paper submitted to the expert group meeting entitled "The role of men and boys in achieving gender equality"*, United Nations Division for the Advancement of Women. Document cited in United Nations Economic and Social Commission for Asia and the Pacific. United Nations Division for the Advancement of Women (DAW), Brasilia, Brazil.
- [8] Juliet M Corbin and Anselm Strauss. 1990. Grounded theory research: Procedures, canons, and evaluative criteria. *Qualitative sociology* 13, 1 (1990), 3–21.
- [9] Pankaj Doke and Anirudha Joshi. 2015. Mobile phone usage by low literate users. In *Proceedings of the 7th Indian Conference on Human-Computer Interaction*. Association for Computing Machinery, New York, NY, United States, Guwahati India, 10–18.
- [10] Ruth Frankenberg. 2004. On unsteady ground: Crafting and engaging in the critical study of whiteness. In *Living spirit, living practice: Poetics, politics, epistemology*. Routledge, London, 104–118.
- [11] Rohit Gupta, Udayan Vidyanta, and Silpa Murali. 2017. Paisa Stick: Tangible Currency Interface for Cashless Transactions. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, United States, Denver Colorado USA, 60–66.
- [12] Geert Hofstede. 2011. Dimensionalizing cultures: The Hofstede model in context. *Online readings in psychology and culture* 2, 1 (2011), 8.
- [13] Samia Ibtasam, Lubna Razaq, Haider W Anwar, Hamid Mehmood, Kushal Shah, Jennifer Webster, Neha Kumar, and Richard Anderson. 2018. Knowledge, access, and decision-making: Women's financial inclusion in Pakistan. In *Proceedings of the 1st ACM SIGCAS Conference on Computing and Sustainable Societies*. Association for Computing Machinery, New York, NY, United States, Menlo Park and San Jose CA USA, 1–12.

- [14] Rahat Jahangir Rony, Syeda Shabnam Khan, Anik Sinha, Anik Saha, and Nova Ahmed. 2021. "COVID has made Everyone Digital and Digitally Independent": Understanding Working Women's DFS and Technology Adoption during COVID Pandemic in Bangladesh. In *Asian CHI Symposium 2021*. Association for Computing Machinery, New York, NY, United State, Yokohama Japan, 202–209.
- [15] Yannis Karamanos, Aurélie Mateos, Caroline Mysiorek, Julien Saint-Pol, and Sylvie Berger. 2019. Progress with colearning. *FEBS Open Bio*. NJ USA: Wiley 9 (2019), 408.
- [16] Louis J Kruger, Rachel F Rodgers, Stephanie J Long, and Alice S Lowy. 2019. Individual interviews or focus groups? Interview format and women's self-disclosure. *International Journal of Social Research Methodology* 22, 3 (2019), 245–255.
- [17] Zhi-Qin Liu, Evgenij Dorozhkin, Nataliia Davydova, and Nadezhda Sadovnikova. 2020. Co-learning as a new model of learning in a digital environment: Learning effectiveness and collaboration. *International Journal of Emerging Technologies in Learning (iJET)* 15, 13 (2020), 34–48.
- [18] Indrani Medhi, Ed Cutrell, and Kentaro Toyama. 2010. It's not just illiteracy. In *Proceedings of the 2010 International Conference on Interaction Design & International Development* (Mumbai, India) (*IHCI'10*). BCS Learning & Development Ltd., Swindon, GBR, 1–10.
- [19] Indrani Medhi, SN Nagasena Gautama, and Kentaro Toyama. 2009. A comparison of mobile money-transfer UIs for non-literate and semi-literate users. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, United States, Boston MA USA, 1741–1750.
- [20] Indrani Medhi, Somani Patnaik, Emma Brunskill, SN Nagasena Gautama, William Thies, and Kentaro Toyama. 2011. Designing mobile interfaces for novice and low-literacy users. *ACM Transactions on Computer-Human Interaction (TOCHI)* 18, 1 (2011), 1–28.
- [21] Miro. 2023. *An Online whiteboard and visual collaboration platform*. Miro. Retrieved December 10, 2023 from <http://miro.com>
- [22] Maryam Mustafa, Noor Mazhar, Ayesha Asghar, Maryem Zafar Usmani, Lubna Razaq, and Richard Anderson. 2019. Digital financial needs of micro-entrepreneur women in Pakistan: is mobile money the answer?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, United States, Glasgow Scotland Uk, 1–12.
- [23] Maryam Mustafa, Kimia Tuz Zaman, Tallal Ahmad, Amna Batool, Masitah Ghazali, and Nova Ahmed. 2021. Religion and Women's Intimate Health: Towards an Inclusive Approach to Healthcare. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, United States, Yokohama Japan, 1–13.
- [24] Pew Reserach. 2023. *Pew Research Center*. Pew Research Center. Retrieved December 10, 2021 from <https://www.pewresearch.org/>
- [25] Rahat Jahangir Rony, Syeda Shabnam Khan, Anik Sinha, Anik Saha, and Nova Ahmed. 2021. 'I Didn't Understand but I Was Determined to Learn': Understanding the Contrast of Using DFS among the Working Women in Bangladesh. *HCI Across Borders (HCIXB)* HCIXB 2021, 6 (2021), 8.
- [26] Rahat Jahangir Rony, Anik Sinha, Shajnuh Amir, Syeda Shabnam Khan, Anik Saha, Ifiti Azad Abeer, and Nova Ahmed. 2022. "I Use YouTube Now in COVID": Understanding Technology Adoption of Indigenous Communities during COVID-19 Pandemic in Bangladesh. In *ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS)*. Association for Computing Machinery, New York, NY, United States, Seattle WA USA, 596–602.
- [27] Muhammad Azam Roomi and Guy Parrott. 2008. Barriers to development and progression of women entrepreneurs in Pakistan. *The Journal of Entrepreneurship* 17, 1 (2008), 59–72.
- [28] Ricarose Roque, Karina Lin, and Richard Liuzzi. 2016. "I'm Not Just a Mom": Parents Developing Multiple Roles in Creative Computing. In *Transforming Learning, Empowering Learners*. Singapore: International Society of the Learning Sciences, Singapore.
- [29] Nithya Sambasivan, Nova Ahmed, Amna Batool, Elie Bursztein, Elizabeth Churchill, Laura Sanely Gaytan-Lugo, Tara Matthews, David Nemer, Kurt Thomas, and Sunny Consolvo. 2019. Toward gender-equitable privacy and security in south asia. *IEEE Security & Privacy* 17, 4 (2019), 71–77.
- [30] Nithya Sambasivan, Amna Batool, Nova Ahmed, Tara Matthews, Kurt Thomas, Laura Sanely Gaytan-Lugo, David Nemer, Elie Bursztein, Elizabeth Churchill, and Sunny Consolvo. 2019. "They Don't Leave Us Alone Anywhere We Go" Gender and Digital Abuse in South Asia. In *proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, United State, Glasgow Scotland Uk, 1–14.
- [31] Nithya Sambasivan, Garen Checkley, Amna Batool, Nova Ahmed, David Nemer, Laura Sanely Gaytan-Lugo, Tara Matthews, Sunny Consolvo, and Elizabeth Churchill. 2018. "Privacy is not for me, it's for those rich women": Performative Privacy Practices on Mobile Phones by Women in South Asia. In *Fourteenth Symposium on Usable Privacy and Security (SOUPS 2018)*. USENIX Association, 2560 Ninth St. Suite 215 Berkeley, CA United States, Baltimore MD USA, 127–142.
- [32] Ayushi Srivastava, Shivani Kapania, Anupriya Tuli, and Pushpendra Singh. 2021. Actionable UI Design Guidelines for Smartphone Applications Inclusive of Low-Literate Users. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–30.
- [33] Sharifa Sultana, François Guimbretière, Phoebe Sengers, and Nicola Dell. 2018. Design within a patriarchal society: Opportunities and challenges in designing for rural women in bangladesh. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, United State, Montreal QC Canada, 1–13.
- [34] Sharifa Sultana, Shaid Hasan, Khandaker Reaz Mahmud, SM Raihanul Alam, and Syed Ishtiaque Ahmed. 2019. 'Shada Baksho' a hardware device to explore the fears of using mobile phones among the rural women of Bangladesh. In *Proceedings of the Tenth International Conference on Information and Communication Technologies and Development*. Association for Computing Machinery, New York, NY, United States, Ahmedabad India, 1–4.
- [35] Udayan Tandon, Lavanya Siri, Apurv Mehra, and Jacki O'Neill. 2019. Designing a financial management smartphone app for users with mixed literacies. In *Proceedings of the Tenth International Conference on Information and Communication Technologies and Development*. Association for Computing Machinery, New York, NY, United States, Ahmedabad India, 1–5.
- [36] Shelley E Taylor, David K Sherman, Heejung S Kim, Johanna Jarcho, Kaori Takagi, and Melissa S Dunagan. 2004. Culture and social support: Who seeks it and why? *Journal of personality and social psychology* 87, 3 (2004), 354.